Data Source

SOURCE SYSTEMS
**Identified source systems*

Database → ITEC, K2, ITOS, and others
D365 F&O → Azure ADLS
CRM / 7Club → Cloud
Caseclub → S3
GI / EV7 → Google Sheets

Other S3
Online Share file
Database on-premise /cloud
Azure Data Lake

Google Sheets

Ingestion

Kinesis Firehouse
Lambda
DMS
Glue

INGESTION
Select the ingestion method based on the source system

DMS → Direct Database Ingestion — supports CDC
Lambda → Lightweight / Small-Scale Ingestion
Glue → Batch Ingestion — Scheduled Jobs
Kinesis Data Firehose → Near Real-Time Ingestion

Malware & PII Scan

Guard Duty
SES

SECURITY GATE

GuardDuty → Threat & Malware Detection
SES → Email Notifications

Glue ETL

ACCESS CONTROL

IAM
Secret Manager
Lake Formation
MFA

ACCESS CONTROL

IAM → Service & resource permissionsLake **Lake Formation** → Table / Column / Row Permissions
Secrets Manager → Passwords & Credentials
MFA → Multi-Factor Authentication

Raw Data

"Raw Source Data"

- Data Consolidation from Multiple Sources
- Structured / Semi-Structured / Unstructured Data eg. image, CSV.
- Historical Data & Versioning
- Batch & Real-Time Ingestion

Ex.Structure
- S3 — Case club, Icare insurance
- Database — sever.... — ITEC, ITOS, K2 } UFUND
- Share file — Google sheet — EV7, GI
- Azure Datalake — D365 — ITEC

Quarantine

Isolate files that fail security scanning

Audit log

Stores data change logs, including INSRET, UPDATE and DELETE

Bronze

"Standardized Data"

- Malware Scan
- Data Transformation
- Convert to Iceberge

Ex.Structure
-Bronze — ITEC, UFUND, COM7, K2, ITOS, ICARE(apple survice), GI, EV7

Silver

"Cleansed Data"

- Data validation
- Data Cleaning
- Missig value Handling
- Deduplication
- Category Mapping / Grouping

Ex.Structure
-Silver — ci.ITEC, ci.UFUND, K2&ITOS, ci.ICARE(apple survice), ci.GI, ci.EV7

Gold

"Business-Ready Data"

- Consolidate Tables
- Map / Join Tables Across BUs
- Aggregation
- Feature Engineering
- Centralize Customer Data

Ex.Structure
-Gold — DIM_Customer, DIM_Product, Transection, Fact_sale, Map_customer

Analytic

Athena
Redshift

Dashboard

Power BI
Quick

Users

Power BI → Existing BI Platform
QuickSight → AWS BI & Dashboard · AI Features

QUERY LAYER

Athena → Direct Query on S3
Redshift → High-Concurrency Query
For Table size > 100 M row or
Query latency > 2 min.

Monitoring

Step function
Cloud Watch
Cloud Trail
Event Bridge
SNS

ORCHESTRATION & AUDIT

Step Functions → Workflow Orchestration · Retry · Error Handling
EventBridge → Event Triggers
CloudWatch → Event Logs · Alarms
CloudTrail → Audit Logging
SNS → Email Alerts & Notifications

SSOT DATA LAKE